Information Decomposition in Marathon Finish-Time Analysis & Prediction

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Math7343 – Applied Statistics

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Analysis Objective - Pacing

<u>Objective</u>

- Quantify the impact of pacing on overall finish time
- Analyze various effects of positive and negative split times
  - Negative Split – Ran second half faster than first half
  - Positive Split – Ran first half faster than second half
  - Even Split – Ran both halves at roughly the same pace

Hypothesis and Method

- **H_0**: Finish time distributions are identical across pacing groups
- **H_a**: At least one pacing group has a significantly different finish time distribution

Finish times are right skewed – Non-parametric tests

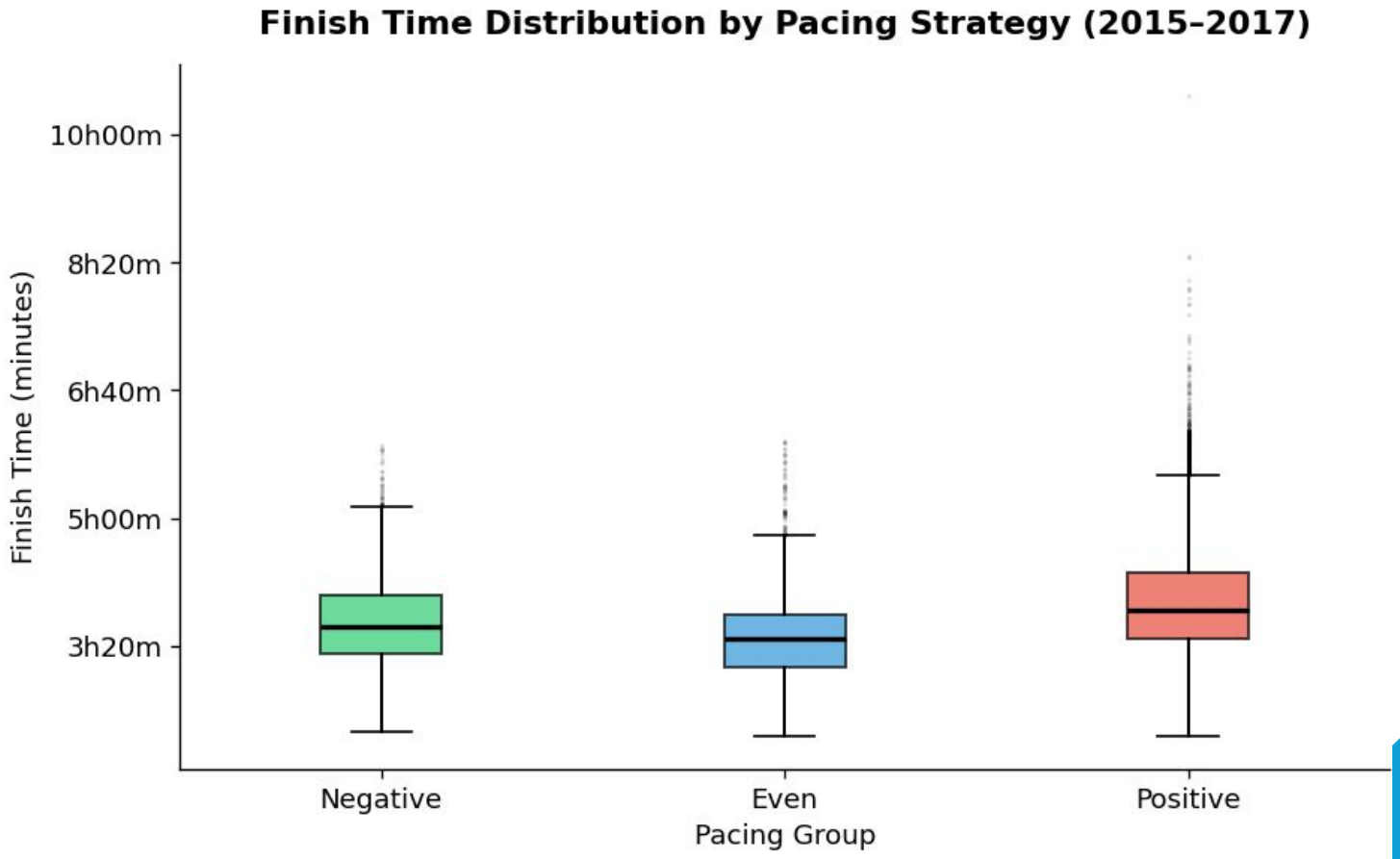
- Kruskal-Wallis
- Dunn's post hoc (with Bonferonni correction)

Kruskal-Wallis / Dunn Results

- **93%** of runners run a positive split
- Small p-value indicated finish time distributions differed between pacing groups

Metric	Value
Runners analyzed	~79,000 (2015–2017)
% Positive split	~93%
% Even split	~4%
% Negative split	~3%
Kruskal-Wallis H	1709.57
p-value	< 0.001
Eta-squared (η^2)	0.019 (1.9% of variance)
All pairwise differences significant?	Yes (Dunn's, Bonferroni)

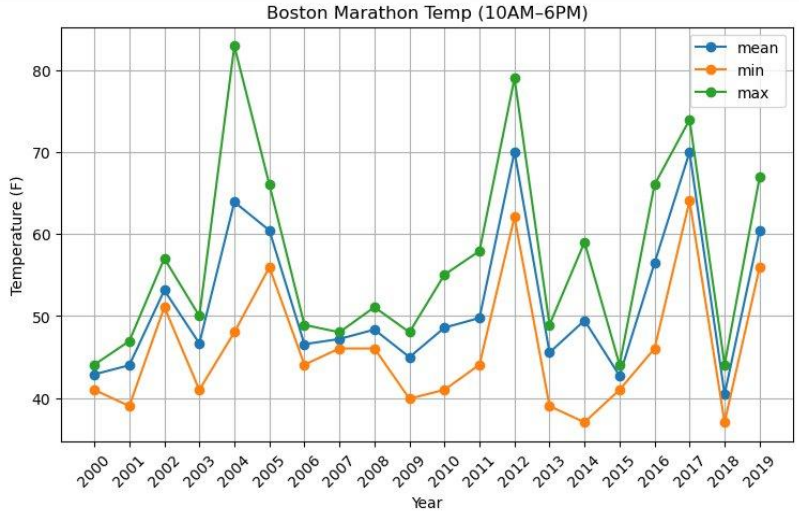
Pacing Distributions

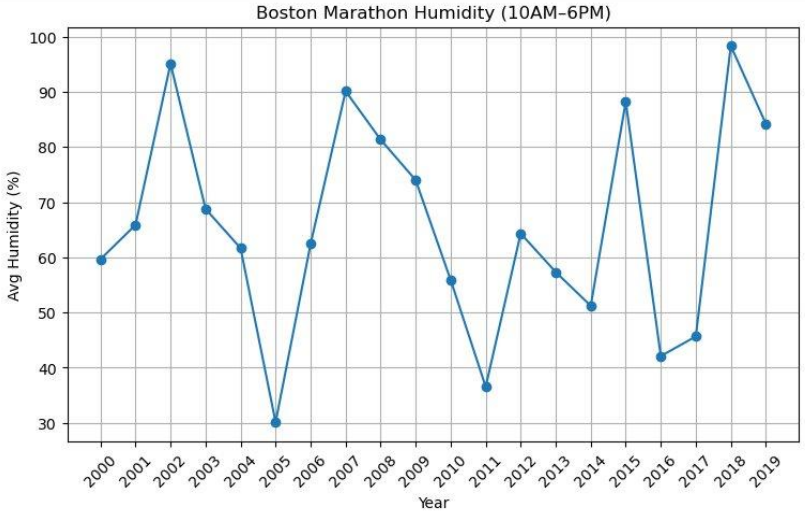


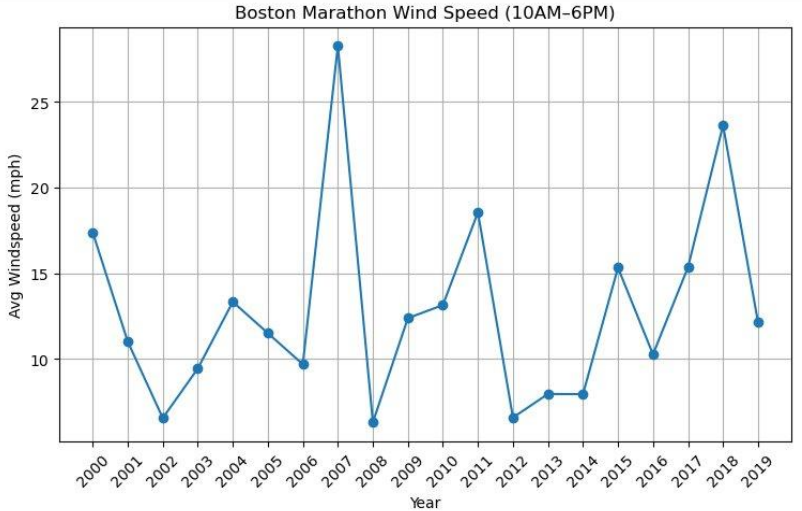
Analysis Objective - Weather

<u>Objective</u>

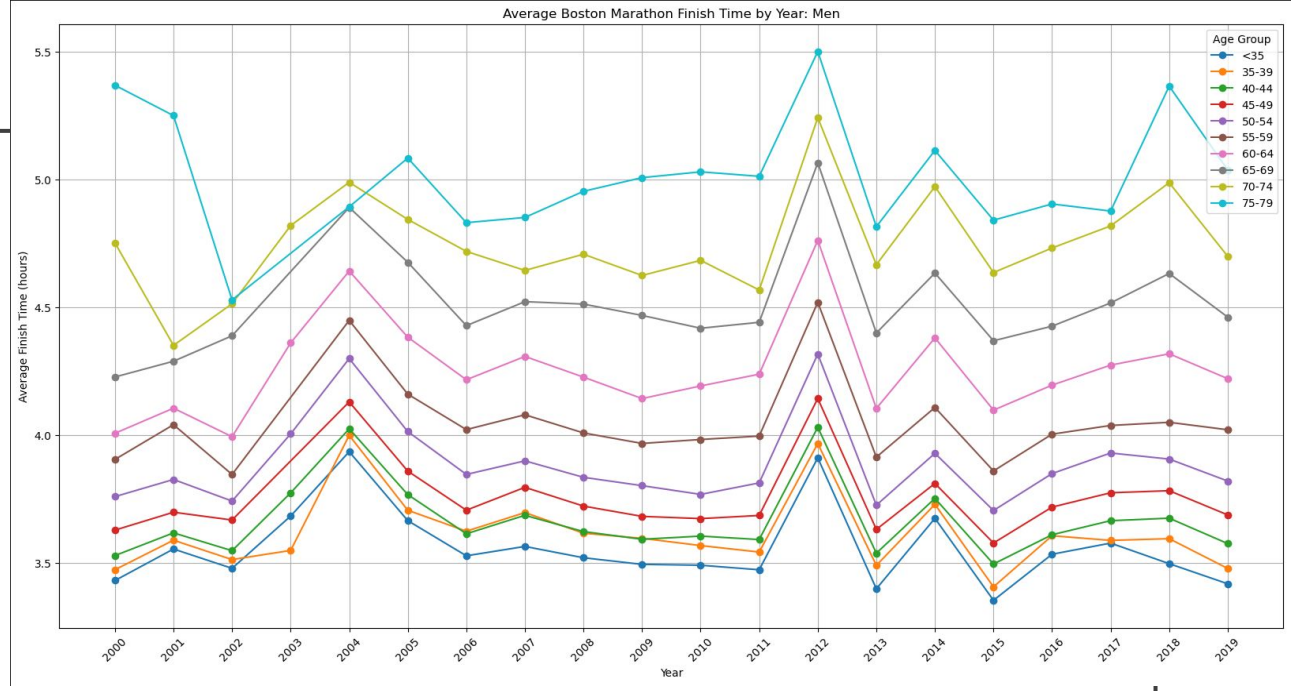
- Quantify impact of external factors (weather) on trends in average finish times each year
  - Analyze age groups separately
  - Temperature
  - Humidity
  - Wind Speed

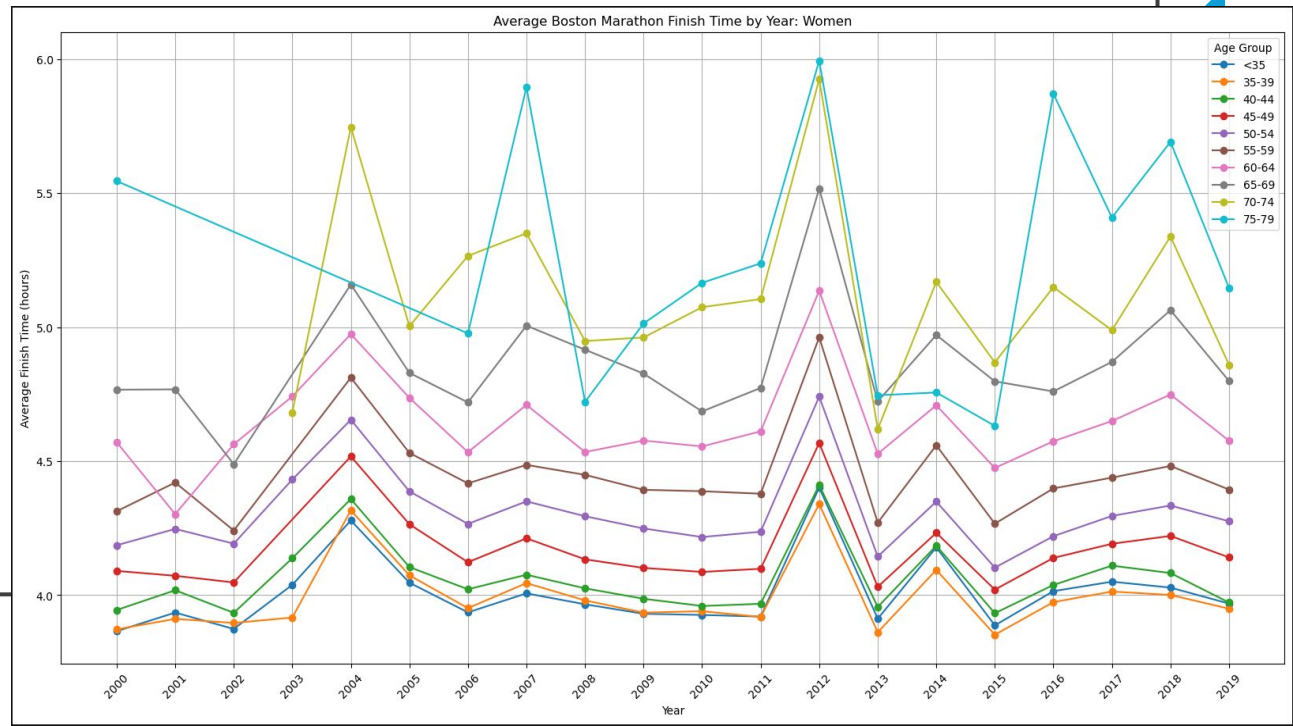






Finish Times by Age Group

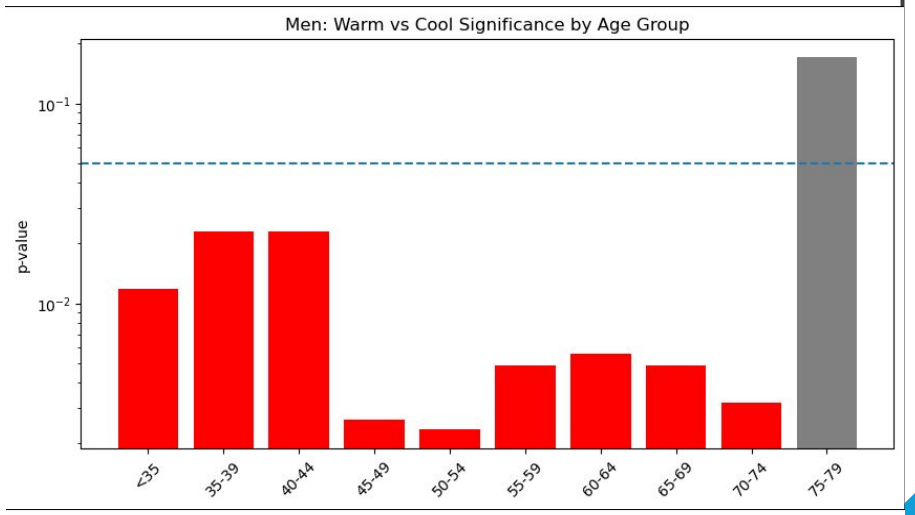


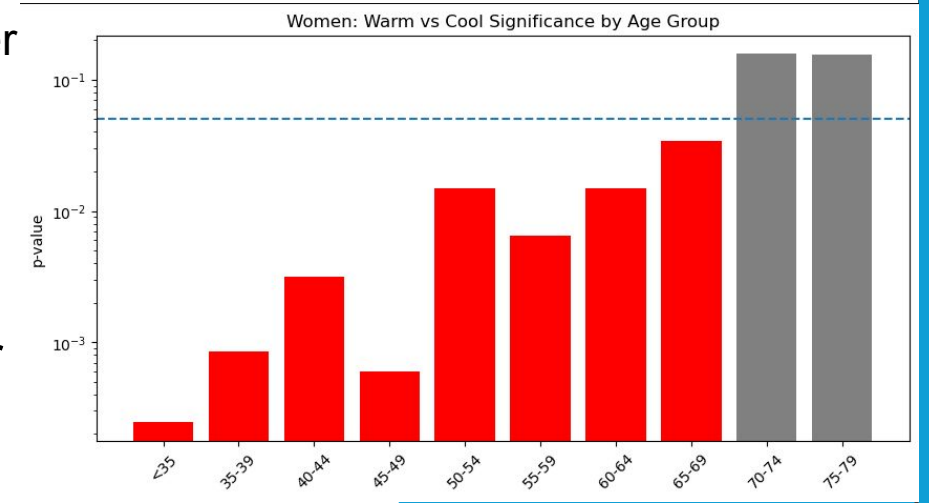


Weather – Test 1

- H_0: Warm weather finish times = cool weather finish times
- H_a: Warm weather finish times > cool weather finish times

- Find general groupings of weather categories
  - Warm vs Cool
  - Humid vs Not Humid
  - Windy vs Not Windy
- Data Assumptions
  - **Not** normally distributed
  - **Small** sample size (groups of averages over large sample population)
- Wilcoxon rank-sum
  - Determines whether one group is significantly larger (slower finish times)
  - **Mann-Whitney U Test**
    - One-sided non-parametric rank-sum test
    - Equal p-values (same conclusion)
- **Only Temperature statistically significant**
- Initial analysis indicates temperature is strongest factor
  - Max temperature stronger than average or median





Weather – Test 2

(One Sided) Spearman

- Based on the strong results from Test 1, find correlation
- Spearman's test determines how much (positive or negative) temperature impacts average finish time
- **Hypothesis**
  - **H_0: $\rho = 0$**
  - **H_a: $\rho > 0$**
- Men:

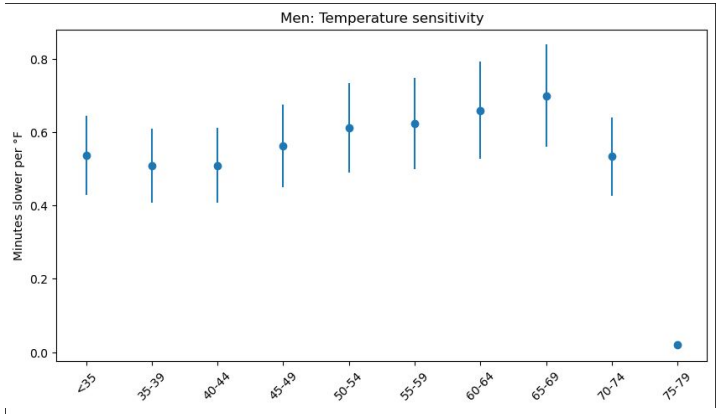
	age	n	rho	p
0	<35	20	0.487751	0.029139
1	35-39	20	0.421411	0.064237
2	40-44	20	0.442519	0.050722
3	45-49	19	0.556096	0.013420
4	50-54	20	0.550322	0.011930
5	55-59	19	0.469866	0.042367
6	60-64	20	0.552584	0.011514
7	65-69	19	0.566655	0.011417
8	70-74	20	0.428196	0.059627
9	75-79	19	0.008799	0.971481

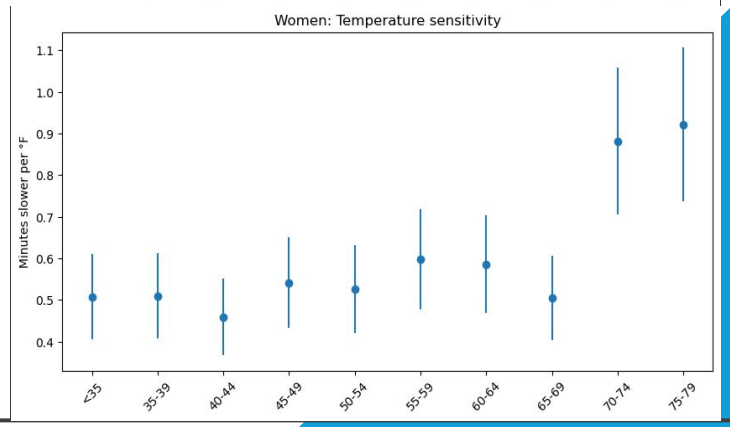
Women:

	age	n	rho	p
0	<35	20	0.617416	0.003726
1	35-39	20	0.609124	0.004362
2	40-44	20	0.520168	0.018719
3	45-49	19	0.583373	0.008744
4	50-54	20	0.473428	0.034992
5	55-59	19	0.439949	0.059437
6	60-64	20	0.462874	0.039863
7	65-69	19	0.268369	0.266606
8	70-74	17	0.235873	0.362081
9	75-79	15	0.229806	0.409977

Results

- Strong positive correlation between higher temperatures and average finish times
- All age groups affected
  - Larger variation and less correlation for 70+ (Men) and 65+ (women)

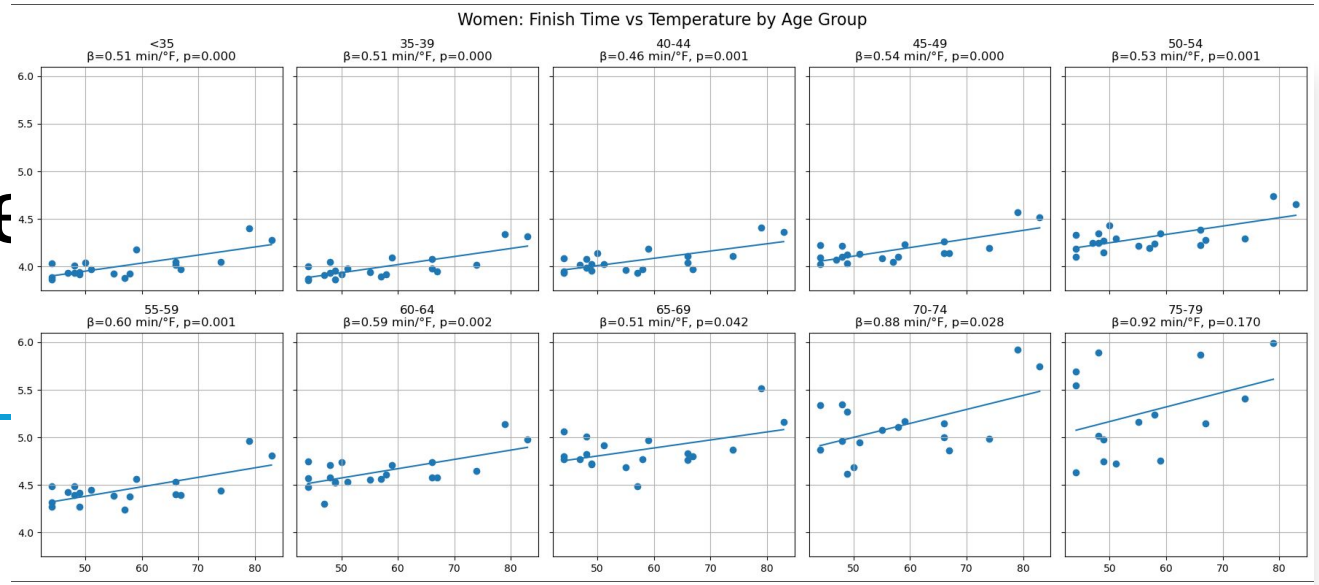


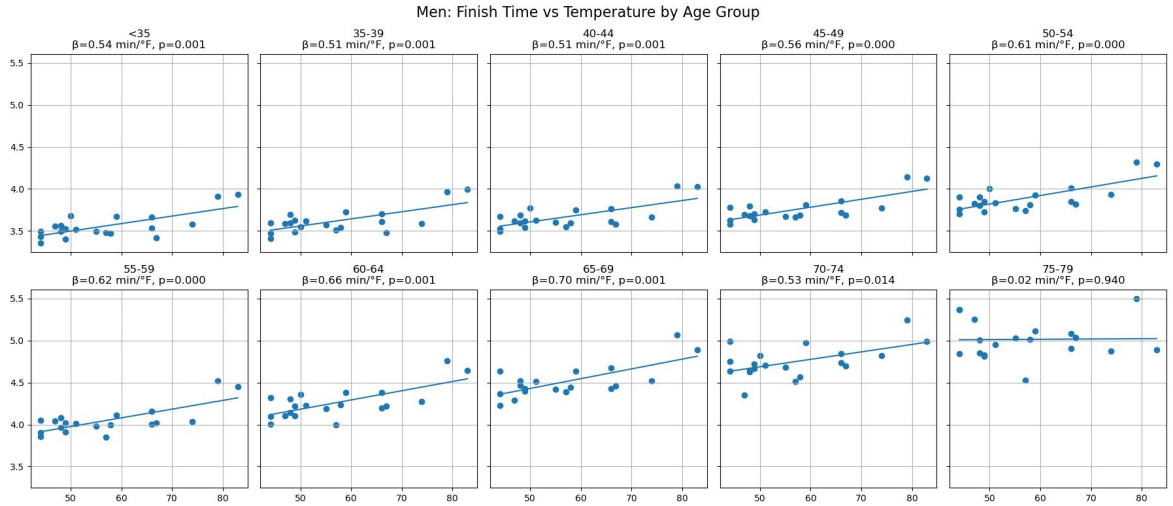


Temperature vs Time Linear Regression

Linear Regression (per age group)

- Y = average finish time (hours)
- X = maximum race-day temperature (°F)
- Y = β0 + β1 X
- **R^2 values ~0.2 - 0.35**
- Temperature explains approximately **20%–35% of the variation** in marathon finish times depending on gender and age:
  - Men – 20-30% for < 75 year olds
  - Women – 25-35% for < 65 year olds
- **>30 sec increase in finish time per ° F**





Prediction Objective

<u>Objective</u>

- Quantify how predictive accuracy evolves as information becomes available
- Decompose accuracy into contributions from:
  - Demographics
  - Prior race history
  - In-race split times

Information Timeline

Information arrices sequentially:

- Registration -> demographics
- Pre-race -> prior history
- In-race -> cumulative splits

Statistical Framing

Prediction Target

Let y_i denote marathon finish time for runner *i*

We compare models along the runner information timeline:

- Pre-race information
- Prior history
- In-race split information

Performance Comparison

$$\Delta RMSE = RMSE_{k-1} - RMSE_k$$

Interpretation

Positive $\Delta RMSE$ means the added information improves prediction. This follows the paper's nested ablation logic, where each model adds one information source at a time.

Evaluation Metrics

Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y}_i)^2}$$

Conformation Prediction Interval

$$[\bar{y}_i - \bar{q}, \bar{y}_i + \bar{q}]$$

With conformal quantile

$$\bar{q} = quantile\left(\left|\bar{y}_i - \hat{y}_i^{(-i)}\right|, \left|\frac{(n+1)(1-\alpha)}{n}\right|\right), \alpha = 0.10$$

Baseline Model

<u>Sample Mean Predictor</u>

$$\hat{y}_i = \bar{y}_{train} = \frac{1}{n}\sum_{j-1}^{n} y_j$$

- Predict every runner with training-set mean
- Serves as the no-information benchmark

Demographic Model

OLS with Demographic Covariates

$$\log(s_i) = \beta_0 + \beta_i a_i + \beta_2 a_i^2 + \beta_3 f_i + \beta_4 t_i + \beta_5 (a_i f_i) + \beta_6 (a_i^2 f_i) + \varepsilon_i$$

Where:

s_i=finish time in seconds

a_i=centered age

f_i=female indicator

t_i=centered year trend

Interpretation:

- Captures nonlinear age-performance trends
- Accounts for gender differences

History-Augmented Model

OLS with Prior History

$$\log(s_i) = x_i^T\beta + \beta_7 \log(1 + k_i) + \beta_8 s_i^{-prior} + \varepsilon_i$$

Where:

k_i=prior Boston appearances

s_i^{-prior}=expanding-window mean of prior finish times

Interpretation:

- Adds the richest pre-race information
- Lets us test whether historical performance materially improves beyond demographics alone
- Expanding-window avoids data leakage
- Log transform captures diminishing returns

Personalization Model

Linear Mixed-Effects Model

$$\log(s_{ij}) = x_{ij}^T\beta + u_{0i} + u_{1i}a_{ij} + \varepsilon_{ij}$$

Where:

u_{0i}=runner-specific random intercept

u_{1i}=runner-specific age slope

Interpretation:

- Accounts for repeated measurements from the same runner
- Tests whether personalization helps before split data arrives

In-Race Models

Ridge Regression at Checkpoints

For each checkpoint c,

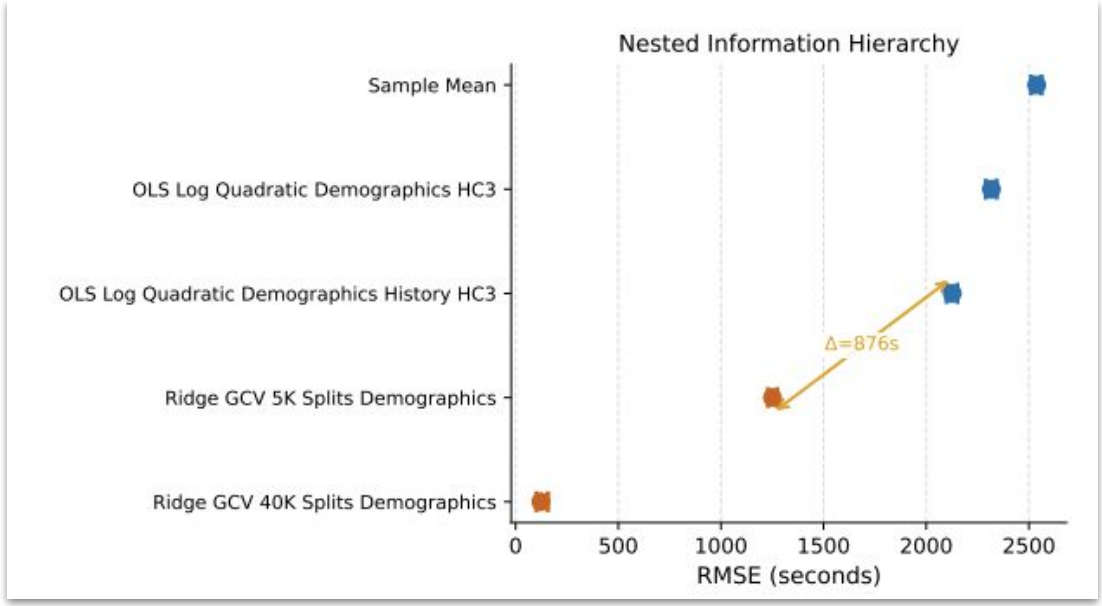
$$\widehat{\beta_c} = \arg\min\{\|y - X_c\beta\|^2 + \lambda_c\|\beta\|^2\}$$

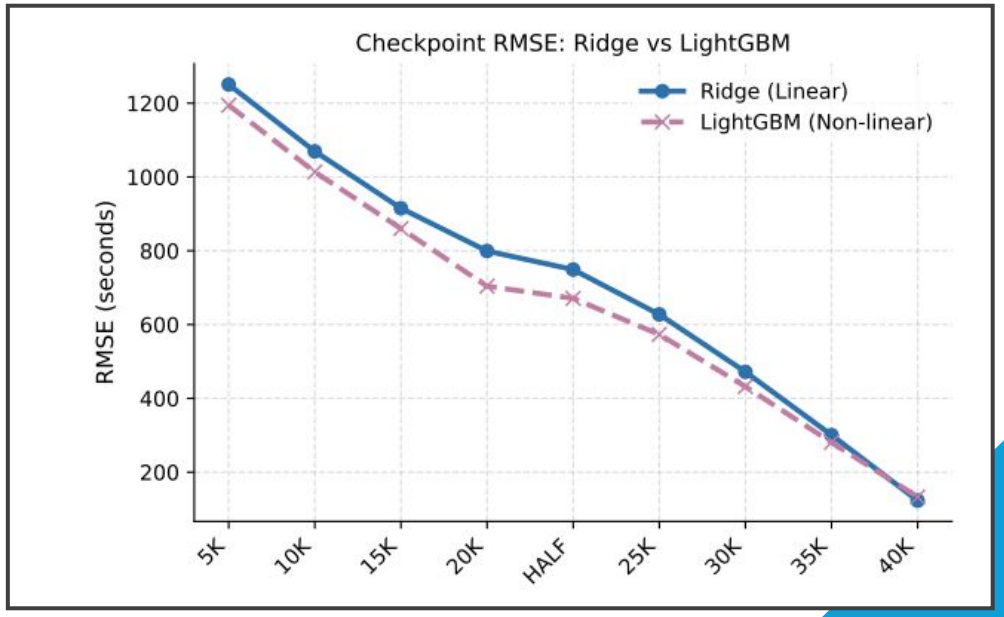
Where:

- X_c includes cumulative split times throughout checkpoint c, plus demographics
- Addresses strong collinearity in cumulative splits

Gradient Boosting (Light GBM)

- Builds a sequence of small decision trees, using gradient descent to iteratively improve
  - Captures interactions and nonlinearities
- Additional features involving ratios, interactions, within-cohort comparisons
  - Segment paces
  - Pace changes
  - Cohort-normalized pace z-scores
- Used as a complexity ceiling - is accuracy limited by the model or the information?


Nested Information Hierarchy
Sample Mean
OLS Log Quadratic Demographics HC3
OLS Log Quadratic Demographics History HC3
Ridge GCV 5K Splits Demographics
Ridge GCV 40K Splits Demographics
Δ=876s
0
500
1000
1500
2000
2500
RMSE (seconds)


Checkpoint RMSE: Ridge vs LightGBM
Ridge (Linear)
LightGBM (Non-linear)
RMSE (seconds)
1200
1000
800
600
400
200
5K
10K
15K
20K
HALF
25K
30K
35K
40K

Main Information Hierarchy Result

Nested Information Hierarchy

- Sample mean: RMSE = 2538.0
- Demographics: RMSE = 2317.3
- History: RMSE = 2126.8
- 5K split: RMSE = 1250.8
- 40K split: 123.0

Main Finding

- The first 5K split reduces RMSE by 876.0 seconds
- That is larger than the combined gain from demographics and history

Statistical Significance of RMSE Improvements

Objective:

- Test whether RMSE differences between adjacent models are statistically significant

Assumptions of standard CI:

- Independent observations
- Approximately normal sampling distribution

Violations in our data:

- Repeated runners → within-runner dependence
- RMSE differences may be skewed due to influential observations

BCa Cluster-Bootstrap Framework

Resampling Procedure:

- Resample runners (clusters) with replacement
- Preserve within-runner dependence

Bootstrap Distribution:

$$\hat{\theta}_b^*, b = 1, \dots, B$$

Where B=1,000 bootstrap samples

- RMSE difference recomputed for each sample
- $\{\hat{\theta}_1^*, \dots, \hat{\theta}_b^*\}$ approximates sampling distribution

Limitations of Percentile Method:

- Assumes symmetry and correct centering
- Not appropriate under skewedness

BCa Adjustments and CI Construction

Boostrap-Based Inference

BCa cluster-bootstrap confidence intervals are used to evaluate RMSE differences between models.

Acceleration(skewedness):

$$a = \frac{\sum_{i-1}^{m} d_i^3}{6\left(\sum_{i-1}^{m} d_i^2\right)^{3/2}}$$

Bias Correction:

$$z_0 = \phi^{-1}\left(clip\left(\frac{\#\{\hat{\theta}_b^* < \hat{\theta}\}}{B}, \frac{1}{B+1}, \frac{B}{B+1}\right)\right)$$

Final Confidence Interval:

$$\left[\hat{\theta}^*_{(Plo)}, \hat{\theta}^*_{(Phi)}\right]$$

Final Takeaway:

95% Bca intervals excluding 0 indicate statistically significant RMSE improvmeents

Conclusion

<u>Key Findings</u>

- Early in-race information dominates pre-race information
- Personalization helps, but only modestly
- Model complexity adds little once information is rich
- Predictive accuracy is limited more by the information frontier than by model class